

Supplementary Material for “Toward a Tractability Frontier for Exact Relevance Certification”

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1 Full Lean Handle Ledger

This supplement provides the complete Lean handle ledger cited by the manuscript. It includes every handle identifier, declaration name, and source module path.

ID	Lean Handle / Source	ID	Lean Handle / Source
FR1	currentFamilies_partition_by_role Paper4dFrontier/Classification.lean	FR2	currentFamilies_partition_counts Paper4dFrontier/Classification.lean
FR3	current_families_have_finite_basis Paper4dFrontier/Classification.lean	FR4	current_positive_interfaces_factor_through_role Paper4dFrontier/Classification.lean
FR5	lifted_families_reduce_to_core Paper4dFrontier/Classification.lean	FR6	degenerateFamilies_complete Paper4dFrontier/Classification.lean
FR7	exists_decisionProblem_realizing_labeling Paper4dFrontier/Realizability.lean	FR8	realizingProblem_decisionEquiv_iff Paper4dFrontier/Realizability.lean
FR13	isOptimal_positiveAffineTransform_iff Paper4dFrontier/FamilyAxioms.lean	FR14	opt_eq_positiveAffineTransform Paper4dFrontier/FamilyAxioms.lean
FR15	isSufficient_positiveAffineTransform_iff Paper4dFrontier/FamilyAxioms.lean	FR16	isRelevant_positiveAffineTransform_iff Paper4dFrontier/FamilyAxioms.lean
FR17	decisionEquiv_relabelActions_iff Paper4dFrontier/FamilyAxioms.lean	FR18	decisionEquiv_relabelStates_iff Paper4dFrontier/FamilyAxioms.lean
FR19	isSufficient_relabelActions_iff Paper4dFrontier/FamilyAxioms.lean	FR20	isSufficient_relabelStates_iff Paper4dFrontier/FamilyAxioms.lean
FR21	allProblems_relabelInvariant Paper4dFrontier/FamilyAxioms.lean	FR22	allProblems_kernelUniversal Paper4dFrontier/FamilyAxioms.lean
FR23	relabelInvariant_not_enough Paper4dFrontier/FamilyAxioms.lean	FR24	optRangeCard_realizingIdentity Paper4dFrontier/FamilyAxioms.lean
FR25	currentFamilies_explicit Paper4dFrontier/Classification.lean	FR26	coreFamilies_explicit Paper4dFrontier/Classification.lean
FR27	liftedFamilies_explicit Paper4dFrontier/Classification.lean	FR28	degenerateFamilies_explicit Paper4dFrontier/Classification.lean
FR29	primitiveMechanisms_explicit Paper4dFrontier/Classification.lean	FR30	isSufficient_iff_of_decisionEquiv_iff Paper4dFrontier/FamilyAxioms.lean
FR31	isRelevant_iff_of_decisionEquiv_iff Paper4dFrontier/FamilyAxioms.lean	FR32	isIrrelevant_iff_of_decisionEquiv_iff Paper4dFrontier/FamilyAxioms.lean
FR33	isMinimalSufficient_iff_of_decisionEquiv_iff Paper4dFrontier/FamilyAxioms.lean	FR34	sufficientSets_eq_of_decisionEquiv_iff Paper4dFrontier/FamilyAxioms.lean
FR35	relevantSet_eq_of_decisionEquiv_iff Paper4dFrontier/FamilyAxioms.lean	FR37	quotientCard_eq_of_decisionEquiv_iff Paper4dFrontier/FamilyAxioms.lean
FR38	quotientCard_realizingIdentity Paper4dFrontier/FamilyAxioms.lean	FR39	realizingProblemQuotientEquivLabelRange_apply_quotientMap Paper4dFrontier/Realizability.lean
FR40	realizingProblem_quotientCard_eq_labelRangeCard Paper4dFrontier/Realizability.lean	FR41	certificationStatistic_eq_of_decisionEquiv_iff Paper4dFrontier/FamilyAxioms.lean

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ID	Lean Handle / Source	ID	Lean Handle / Source
FR42	sufficientSetCount_eq_of_decisionEquiv_iff Paper4dFrontier/FamilyAxioms.lean	FR43	minimalSufficientSetCount_eq_of_decisionEquiv_iff Paper4dFrontier/FamilyAxioms.lean
FR44	relevantCoordCount_eq_of_decisionEquiv_iff Paper4dFrontier/FamilyAxioms.lean	FR45	setoidRealizingProblem_decisionEquiv_iff Paper4dFrontier/Realizability.lean
FR46	setoidRealizingProblemQuotientEquivSetoidQuotient_apply_quotientMap Paper4dFrontier/Realizability.lean	FR47	exists_decisionProblem_realizing_setoid Paper4dFrontier/Realizability.lean
FR48	isSufficient_iff_agreementRel_le_decisionEquiv Paper4dFrontier/FamilyAxioms.lean	FR49	isIrrelevant_iff_sufficient_erase Paper4dFrontier/FamilyAxioms.lean
FR50	isRelevant_iff_not_sufficient_erase Paper4dFrontier/FamilyAxioms.lean	FR51	quotientEquiv_of_decisionEquiv_iff_apply_quotientMap Paper4dFrontier/FamilyAxioms.lean
FR52	empty_sufficient_of_constant_opt Paper4dFrontier/ClosureLaws.lean	FR53	irrelevant_of_constant_opt Paper4dFrontier/ClosureLaws.lean
FR54	decisionEquiv_duplicateAction_iff Paper4dFrontier/ClosureLaws.lean	FR55	isSufficient_duplicateAction_iff Paper4dFrontier/ClosureLaws.lean
FR56	isRelevant_duplicateAction_iff Paper4dFrontier/ClosureLaws.lean	FR66	booleanCube_card Paper4dFrontier/EnumerationBounds.lean
FR67	exists_booleanCube_larger_than_monomial Paper4dFrontier/EnumerationBounds.lean	FR68	decisionEquiv_addDuplicateState_iff Paper4dFrontier/ClosureLaws.lean
FR69	isSufficient_addDuplicateState_iff Paper4dFrontier/ClosureLaws.lean	FR70	isRelevant_addDuplicateState_iff Paper4dFrontier/ClosureLaws.lean
FR71	addDuplicateStateQuotientEquivOriginal_apply_quotientMap Paper4dFrontier/ClosureLaws.lean	FR72	some_mem_opt_addDuplicateAction_iff Paper4dFrontier/ClosureLaws.lean
FR73	none_mem_opt_addDuplicateAction_iff Paper4dFrontier/ClosureLaws.lean	FR100	parentTree_realTreewidth_one Paper4dFrontier/ParentTreewidth.lean
FR101	low_rank_only_witness Paper4dFrontier/LowRankOnlyWitness.lean	FR105	separable_only_witness Paper4dFrontier/SeparableOnlyWitness.lean
FR106	bounded_actions_only_witness Paper4dFrontier/BoundedActionsOnlyWitness2.lean	FR108	all_independent_core_mechanisms_nondegenerate Paper4dFrontier/Block2Progress.lean
FR110	parentTreeStructured_implies_treeStructured Paper4dFrontier/ParentTreewidth.lean	FR111	treeStructured_and_unique_parent_iff_parentTreeStructured Paper4dFrontier/ParentTreewidth.lean
FR112	bounded_treewidth_only_witness Paper4dFrontier/BoundedTreewidthOnlyWitness.lean	FR113	symmetry_only_witness Paper4dFrontier/SymmetryOnlyWitness.lean
FR114	weak_tree_structured_not_width_one Paper4dFrontier/LegacyTreeGap.lean	FR115	tree_structure_hardness_bridge Paper4dFrontier/TreeStructureHardnessBridge.lean
FR118	pairCrossDifference_eq_binaryCrossDifference_of_lt Paper4dFrontier/BinaryPairwiseDichotomy.lean	FR119	pairwise_zero_crossDifference_unaryDecomposition Paper4dFrontier/BinaryPairwiseDichotomy.lean
FR120	binary_pairwise_symmetry_dichotomy Paper4dFrontier/BinaryPairwiseDichotomy.lean	FR121	actionGapCrossDifference_eq_binaryCrossDifference_of_lt Paper4dFrontier/DecisionRelevantPairwiseDichotomy.lean
FR122	decisionRelevant_zero_actionGap_implies_unaryReduction Paper4dFrontier/DecisionRelevantPairwiseDichotomy.lean	FR123	binary_pairwise_symmetry_decision_relevant_dichotomy Paper4dFrontier/DecisionRelevantPairwiseDichotomy.lean
FR124	block6_genuine_interaction_dichotomy_obstruction Paper4dFrontier/Block6Obstruction.lean	FR125	decisionRelevantInteractionGraph_addActionOffset Paper4dFrontier/DecisionRelevantPairwiseDichotomy.lean
FR126	action_offset_can_force_constant_optimizer Paper4dFrontier/Block6Obstruction.lean	FR127	block6_action_offset_obstruction Paper4dFrontier/Block6Obstruction.lean
FR128	offsetNormalizedDecisionRelevantInteractionGraph_wellDefined Paper4dFrontier/DecisionRelevantPairwiseDichotomy.lean	FR129	block6_offset_normalized_obstruction Paper4dFrontier/Block6Obstruction.lean
FR130	neverOptimalGhost_decisionRelevantGraph_eq_top Paper4dFrontier/Block6Obstruction.lean	FR131	neverOptimalGhost_supportedGraph_eq_bot Paper4dFrontier/Block6Obstruction.lean
FR132	support_filtering_removes_known_block6_obstructions Paper4dFrontier/Block6Obstruction.lean	FR133	block6_ghost_action_obstruction Paper4dFrontier/Block6Obstruction.lean

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ID	Lean Handle / Source	ID	Lean Handle / Source
FR134	marginMasking_supportedGraph_eq_top Paper4dFrontier/Block6Obstruction.lean	FR135	marginMasking_zero_sufficient Paper4dFrontier/Block6Obstruction.lean
FR136	block6_optimizer_supported_obstruction Paper4dFrontier/Block6Obstruction.lean	FR137	MarginBounded Paper4dFrontier/DecisionRelevantPairwiseDichotomy.lean
FR138	dominantPair_marginBounded Paper4dFrontier/Block6Obstruction.lean	FR139	dominantPair_supportedGraph_eq_top Paper4dFrontier/Block6Obstruction.lean
FR140	block6_margin_bounded_obstruction Paper4dFrontier/Block6Obstruction.lean	FR145	not_collapseLandscapeInfinity_of_oracle_ predicate Paper4dFrontier/MetaCharacterization.lean
FR176	DecisionQuotient.DecisionProblem.quotient_is_ coarsest paper4/DecisionQuotient/Quotient.lean	FR177	DecisionQuotient.DecisionProblem.quotient_has_ unique_factorization paper4/DecisionQuotient/Quotient.lean
FR178	DecisionQuotient.DecisionProblem.srank_eq_ relevant_card paper4/DecisionQuotient/Tractability/ StructuralRank.lean	FR179	DecisionQuotient.quotientEntropy_le_srank_ binary paper4/DecisionQuotient/Information.lean
FR180	DecisionQuotient.Statistics.fisherMatrix_rank_ eq_srank paper4/DecisionQuotient/Statistics/ FisherInformation.lean	FR181	DecisionQuotient.Information.compression_ below_srank_fails paper4/DecisionQuotient/Information/ RDSrank.lean
FR182	DecisionQuotient.Information.srank_bits_ sufficient paper4/DecisionQuotient/Information/ RDSrank.lean	FR183	DecisionQuotient.Physics.single_future_zero_ cost paper4/DecisionQuotient/Physics/ WassersteinIntegrity.lean
FR184	DecisionQuotient.Physics.transportCost_pos_of_ offDiag paper4/DecisionQuotient/Physics/ WassersteinIntegrity.lean	FR185	no_closureInvariant_predicate_of_orbit_gap Paper4dFrontier/ObstructionPredicateCandidates.lean
FR186	no_admissibleNormalizationPredicate_decides_ dominantPair Paper4dFrontier/ObstructionPredicateCandidates.lean	FR187	no_admissibleNormalizationPredicate_decides_ marginBounded Paper4dFrontier/ObstructionPredicateCandidates.lean
FR188	no_admissibleNormalizationPredicate_decides_ ghostActionTwoPairCrossOne Paper4dFrontier/ObstructionPredicateCandidates.lean	FR189	no_admissibleNormalizationPredicate_decides_ offsetActionZeroPairCrossOne Paper4dFrontier/ObstructionPredicateCandidates.lean
FR190	admissibleCollapseLandscapeInfinity_full_paper Paper4dFrontier/ObstructionPredicateCandidates.lean		

2 Claim-to-Lean Handle Mapping

This section maps each paper claim to its corresponding Lean formalization.

Paper claim	Lean handle
Corollary 6.6: Constant-Optimizer Collapse Forces Trivial Certification	FR52, FR53
Corollary 2.3: Finite Basis for the Current Positive Landscape	FR3, FR29
Corollary 5.4: Realizability Alone Cannot Be the Frontier	FR7, FR8
Corollary 6.3: Relevance Is Failure of Erased Sufficiency	FR49, FR50
Corollary 6.4: Certification Statistics Factor Through the Quotient Relation	FR41, FR43, FR44, FR42
Corollary 6.2: Sufficiency Is Relation Refinement	FR48
Proposition 6.8: Positive Affine Utility Reparameterizations Are Invisible	FR13, FR16, FR15, FR14
Proposition 4.1: Binary Pairwise Symmetry Dichotomy	FR120, FR118, FR119
Proposition 4.2: Decision-Relevant Binary Pairwise Dichotomy	FR121, FR123, FR124, FR122
Proposition 3.2: Degenerate Positive Cases	FR1, FR6

Paper claim	Lean handle
Proposition 6.1: Exact Certification Depends Only on the Decision Quotient Relation	FR32, FR33, FR31, FR30, FR37, FR51, FR35, FR34
Proposition 6.9: Duplicate Actions and Duplicate States Preserve Certification	FR71, FR68, FR54, FR70, FR56, FR69, FR55, FR73, FR72
Proposition 6.7: Explicit Enumeration Is Parameter-Dependent, Not Structural	FR66, FR67
Proposition 2.4: Explicit Primitive Basis	FR29
Proposition 2.2: Explicit Inventory	FR26, FR25, FR3, FR28, FR27
Proposition 2.5: All Independent Core Mechanisms Are Nondegenerate	FR108, FR106, FR112, FR101, FR105, FR113
Proposition 2.9: Current Positive Interfaces Factor Through Role Classification	FR4
Proposition 6.12: Invariance Alone Is Too Weak	FR22, FR21, FR23
Proposition 2.6: Weak Tree Ordering Does Not Imply Width One	FR115, FR114
Proposition 3.1: Lifted Cases Introduce No New Primitive	FR5
Proposition 6.14: Unrestricted Predicates Trivialize Exact Characterization	FR145
Proposition 4.3: Orbit-Gap Template	FR185
Proposition 2.8: Parent-Tree Structure Gives Width-One Decompositions	FR110, FR100, FR111
Proposition 6.11: Quotient Size Is Unbounded Under Realizability	FR24, FR38
Proposition 5.3: The Realized Quotient Is Exactly the Label Range	FR39, FR40
Proposition 6.10: Relabeling Invariance Is Forced by Exact Certification	FR17, FR18, FR19, FR20
Proposition 5.2: Every Equivalence Relation Is Optimizer-Realizable	FR47, FR46, FR45
Proposition 6.13: Independent Summary Frameworks Converge on the Same Structural Core	FR178, FR181, FR182, FR180, FR179
Proposition 2.7: Cyclic Dependencies Recover Hardness	FR110, FR100, FR111, FR115
Proposition 6.5: Zero-Distortion Summaries Refine the Optimizer Quotient	FR177, FR176
Theorem 4.8: Admissible Collapse Across the Four Obstruction Families	FR190
Theorem 2.1: Explicit Partition of the Current Positive Landscape	FR26, FR25, FR1, FR2, FR28, FR27
Theorem 4.4: Dominant-Pair Admissible No-Go	FR186
Theorem 4.6: Ghost-Action Admissible No-Go	FR188
Theorem 5.1: Every Labeling Kernel Is Optimizer-Realizable	FR7, FR8
Theorem 4.5: Margin-Masking Admissible No-Go	FR187
Theorem 4.7: Offset Admissible No-Go	FR189

Auto summary: mapped 37/37 (full=37, derived=0, unmapped=0).